Volumes by the Disc and Washer Methods

Section 1: The Disc Method Around Coordinate Axes

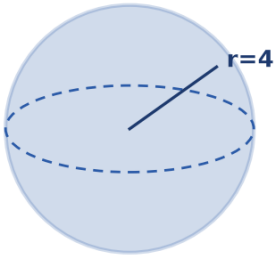
1 Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ about the x-axis.

a. 8π

b. 16π

c. 4π

d. $\frac{16\pi}{3}$



2 Find the volume when the region bounded by $y = x^2, x = 0,$ and $y = 4$ is revolved about the y-axis.

a. 8π

b. 16π

c. 4π

d. $\frac{32\pi}{5}$

3 What is the cross-sectional shape of a solid formed by the disc method?

a. A solid circular disk of area πR^2

b. A circular ring (washer)

c. A rectangle

d. A cylindrical shell

Section 2: The Washer Method Around Coordinate Axes

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4 The region enclosed by $y = x$ and $y = x^2$ is revolved about the x-axis. What is the resulting volume?

a. $\frac{2\pi}{15}$

b. $\frac{\pi}{6}$

c. $\frac{\pi}{15}$

d. $\frac{4\pi}{15}$

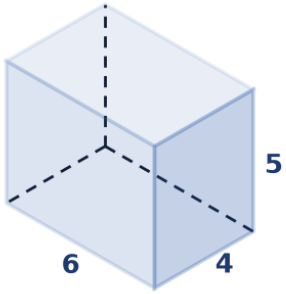
5 Write the integral for the volume when the region between $y = 2x$ and $y = x^2$ is revolved about the x-axis.

a. $\pi\int_0^2 (4x^2 - x^4)dx$

b. $\pi\int_0^2 (2x - x^2)^2 dx$

c. $\pi\int_0^2 (x^4 - 4x^2)dx$

d. $2\pi\int_0^2 (2x - x^2)dx$



6 Why is $\pi\int (R - r)^2 dx$ INCORRECT for the washer method?

a. The area of a washer is $\pi R^2 - \pi r^2$, which is NOT equal to $\pi(R - r)^2$

b. It omits the factor of 2

c. It is only valid for shells

d. It produces negative volume

Section 3: Revolutions Around Non-Axis Horizontal Lines

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7 The region bounded by $y = x^2$ and $y = 4$ is revolved about the line $y = 4$. What is the outer radius $R(x)$?

a. $R(x) = 4 - x^2$

b. $R(x) = x^2$

c. $R(x) = 4$

d. $R(x) = x^2 - 4$

8 Find the volume in the above problem revolved about $y = 4$.

a. $\frac{256\pi}{15}$

b. $\frac{128\pi}{15}$

c. $\frac{64\pi}{5}$

d. 16π

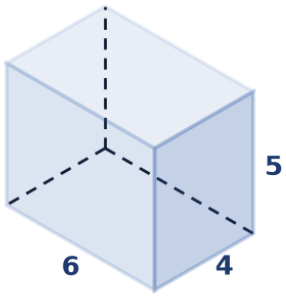
9 The region between $y = \sqrt{x}, y = 0,$ and $x = 4$ is revolved about the line $y = -1$. What are the washer radii?

a. $R(x) = \sqrt{x} + 1$ and $r(x) = 1$

b. $R(x) = \sqrt{x} - 1$ and $r(x) = -1$

c. $R(x) = \sqrt{x}$ and $r(x) = 1$

d. $R(x) = 4$ and $r(x) = \sqrt{x}$



Section 4: Volumes with Known Cross-Sections

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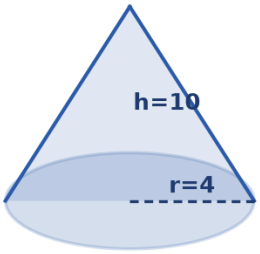
10 The base of a solid is the region enclosed by $y = 1 - x^2$ and the x-axis. Cross-sections perpendicular to the x-axis are squares. What is the volume?

a. $\frac{16}{15}$

b. $\frac{8}{15}$

c. $\frac{4}{3}$

d. $\frac{32}{15}$



11 If cross-sections perpendicular to the x-axis are semicircles with diameter on the base, what is the cross-sectional area formula?

a. $A(x) = \frac{\pi}{8}[s(x)]^2$

b. $A(x) = \frac{\pi}{2}[s(x)]^2$

c. $A(x) = \frac{\pi}{4}[s(x)]^2$

d. $A(x) = \pi[s(x)]^2$

12 If cross-sections are equilateral triangles with side length $s(x)$, what is the cross-sectional area formula?

a. $A(x) = \frac{\sqrt{3}}{4}[s(x)]^2$

b. $A(x) = \frac{1}{2}[s(x)]^2$

c. $A(x) = \frac{\sqrt{3}}{2}[s(x)]^2$

d. $A(x) = \frac{1}{4}[s(x)]^2$

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